Data Science Practicum 1, Proposal

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Abstract

This project is about looking at asset rental and lease agreement data to make business decisions, manage costs, and make things run more smoothly. The data i have includes information about contracts, rental payments, how many assets we have, who made them, who we buy from, how much they cost, tags for maintenance, where they are, and taxes on rentals. I want to find patterns in how assets are used, what we spend on rentals, how well vendors do, and how contracts are handled. I will use things like looking at the data, cleaning it up, making pictures of it, trying to predict what will happen, and making reports to help us understand what is going on. This will help organizations cut costs they do not need and make their asset management better.

I will also look at how rental payments, asset costs, contract terms, manufacturers, and locations are connected. I might use machine learning to guess what rental expenses will be and find contracts that cost much or are not used well. I will also think about things like keeping data private, making sure it is accurate, and being transparent while we are analyzing the data.

What we hope to get information from this project is a way to make decisions based on data that will help with planning and tracking assets. This project will give us ideas that organizations can use to manage the equipment and technology assets they lease. This project will also help us get better at using Python,Powerpoint slides making pictures of data, machine learning, and business analytics.

I. Introduction/Background

This Organizations are using technology leasing more and more to manage their spending, keep their equipment up-to-date, and have flexibility with their money. This company's managing all their IT leases can be really complicated because they have many different assets, contracts, vendors, and locations to keep track of. This organization, where I work, is a global technology company based in Lexington, Kentucky. They have an IT hardware lease portfolio. This organization gets reports from their leasing vendor that have clear information about each asset.

1. The reports include information like service tags
2. Contract identifiers
3. Commencement and termination dates
4. Rental amounts
5. Asset costs
6. Physical locations
7. Equipment descriptions

The January 2025 report shows this organization has 1,724 assets across 29 contracts. Most of these assets are in Kentucky with a few in Colorado. Now this organization relies on people manually reviewing these reports to manage their leases. However, this approach is not good at finding problems, optimizing lease renegotiations, or detecting billing errors. This project aims to use data science to turn this organization's lease data into a strategic asset. By doing so, the organization can better manage its leases and make more informed decisions. Lease data can help companies make choices about their technology investments. This organization wants to use data to improve its lease management process.

II. Problem Statement

Organizations that lease assets often do not have a system to track rental costs, contract performance, and asset usage. This can cause operational costs, duplicate assets, poor vendor management, and inaccurate financial predictions.

By analyzing lease agreement data can help organizations identify cost-saving opportunities and improve asset management

The goal of this project is to analyze asset rental agreement data and provide insights that help organizations:

1. Analyze total rental spending.
2. Identify high-cost assets and lease contracts.
3. Evaluate vendor and manufacturer performance.
4. Track asset locations and distribution.
5. Forecast future rental expenses using predictive analytics.
6. Support better financial planning and operational decision-making

The project aims to turn raw rental agreement data into useful business information using data science and machine learning techniques.

III. Related Work

I noticed that previous management studies have shown that using methods and visual tools can help companies work better and save money. Companies often use systems to keep an eye on contracts and vendors and the things they have in stock. Research has found that computers can help predict costs and risks when it comes to taking care of assets. People also use tools to look at data and make better decisions about what to buy. Most of the companies use dashboard systems like Power BI and Tableau to keep an eye on how things are going with leasing and operations. They use methods like looking at numbers and finding patterns to help with financial decisions and making the most of their assets. This project takes these ideas. Applies them to real data about renting assets. It uses data to understand what is going on, make predictions, and create reports to help companies plan better and make decisions about money and operations.

IV. Methodology / Approach

The project will follow a structured process for the data science lifecycle:

V. Data Analysis

The uploaded dataset contains 1,727 rows and 33 columns related to asset rental agreements and contract management. The dataset includes financial, vendor, asset, and contract information used for lease asset analysis.

Key Dataset Attributes

- Service Tag
- Account Name
- Contract Number
- Rental Payment
- Asset Quantity
- Vendor Name
- Asset Description
- Manufacturer
- Model Number
- Asset Cost

- Asset Rental Amount
- Asset Rental Tax

Analysis Focus Areas

- Missing data visualization
- Asset distribution by state
- Rental amount distribution
- Asset cost vs rental amount relationship
- Vendor and manufacturer cost analysis
- Distribution of Asset Rental Amount
- Asset Cost vs. Monthly Rental

VI. Expected Outcomes

The expected outcomes of the practicum include:

- Better understanding of asset rental costs and lease agreements.
- Identify high-cost assets, vendors, and manufacturers.
- Detect unusual rental charges and pricing outliers.
- Identify the key factors that influence rental costs.
- Support cost reduction and better financial planning.

This project is expected to support better operational planning and financial decision-making.

VII. Timeline

TABLE I
PROJECT TIMELINE

Phase	Activities	Duration
Phase 1	The Data Collection and Understanding	Week 1
Phase 2	The Data Cleaning and Preparation	Week 2
Phase 3	The Exploratory Data Analysis	Week 3
Phase 4	The Feature Engineering and Modeling	Week 4
Phase 5	Weekly Project Progress	Week 5
Phase 6	Project Development and Weekly Update	Week 6
Phase 7	Project Development and Practice Presentation	Week 7
Phase 8	Final Presentations and Submissions	Week 8

VIII. Conclusion

This project focuses on using data science techniques to analyze asset rental agreement data and improve business operations, financial planning, and contract management. In this project, I analyzed 1,727 lease assets and 29 contracts.

The project will use data analysis, predictive modeling, and visualization methods to provide insights into rental spending, vendor performance, and asset utilization.

It will also help develop practical skills in Python and machine learning. This project makes operations more efficient, cuts down on costs, and helps make decisions about managing lease assets.

The goal is to improve operations, save money, and ensure proper handling of lease assets. This will help in making choices about lease assets.

Overall, this practicum highlights the value of data science in enterprise asset management and shows how data-driven solutions can help organizations improve performance and optimize financial operations.

[1] [2] [3] [4] [5] [6] [7]

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